ECO735: REPLICATION EXERCISE

INSTRUCTIONS

1. This exercise is based on the paper **"Worms: Identifying Impacts On Education And Health In the Presence Of Treatment Externalitites" by Edward Miguel & Michael Kremer (*Econometrica, Vol.72, No.1, January 2004)***

2. Part 1 of the exercise involves providing explicit answers to the questions. This should be typed up and submitted in pdf form. You can use either LATEX or Word for this. **ONLY** pdf format will be graded. It should NOT be a pdf of a scanned picture of handwritten material.

3. You can use **ONLY** Stata or R for coding. Submit your codes in a Word File. If it is not in a word file format, your assignment will not be graded.

4. I am attaching samples of how your codes and solution file format should look. Make sure your name and roll number is clearly written on both the solution and the codes.

5. Both the PDF and code files are **NECESSARY** to consider it as a complete submission. In the absence of one or the other, the exercise will be awarded 0 marks.

6. **PLAGIARISM** of no kind will be tolerated (both AI and/or other). In such cases **0** marks will be awarded for the assignment and there shall be no opportunity provided to explain your stance.

7. The required datasets, data user guide, and code book is uploaded here.

QUESTIONS

1. Using the provided dataset, replicate Table 1 ("1988 Average Pupil and School Characteristics, Pre-Treatment") from the paper as closely as possible. **(8)**

   (a) Add an additional Column (7) reporting the mean difference in characteristics between Group 1 and Group 2, along with the corresponding t-test for equality of means.

   (b) Report p-values in parentheses below each mean difference.

   (c) Indicate statistical significance using the conventional notation:
   $***p < 0.01, **p < 0.05, *p < 0.10$.

   (d) Based on your results, comment on whether pre-treatment balance across treatment groups appears satisfactory.

2. Using the provided dataset, replicate Table 8 ("School Participation, School-Level Data") from the paper as closely as possible for the main treatment specification. **(12)**

   (a) Clearly state the estimation equation used.

   (b) Interpret the main coefficient estimates, focusing on:

      i. magnitude,
      ii. statistical significance,
      iii. economic significance,
      iv. implications for school participation.